Characterization of Positive (Semi-)Definite Matrices via Eigenvalues

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1 Definition and Statement

A **symmetric matrix** $A \in \mathbb{R}^{n \times n}$ is said to be:

- **Positive semi-definite (PSD)** if for all $x \in \mathbb{R}^n$, $x^T A x \geq 0$.
- **Positive definite (PD)** if for all $x \neq 0$, $x^T A x > 0$.

A symmetric matrix is positive semi-definite if and only if all of its eigenvalues are nonnegative, and positive definite if and only if all of its eigenvalues are positive. This equivalence holds for symmetric (real) matrices or Hermitian (complex) matrices.

2 Proof Sketch

Let $A \in \mathbb{R}^{n \times n}$ be symmetric. By the **spectral theorem**, there exists an orthogonal matrix Q and a diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ such that

$$A = Q \Lambda Q^T.$$

For any nonzero vector $x \in \mathbb{R}^n$, we have

$$x^T A x = (Q^T x)^T \Lambda (Q^T x) = \sum_{i=1}^{n} \lambda_i y_i^2, \quad \text{where } y = Q^T x.$$

Hence:

- If all $\lambda_i \geq 0$, then $x^T A x \geq 0$ for all x, so A is PSD.
- If all $\lambda_i > 0$, then $x^T A x > 0$ for all $x \neq 0$, so A is PD.

Conversely, if $x^T A x \geq 0$ for all x, take $x = Q e_i$ to obtain $\lambda_i \geq 0$.

3 Important Caveat (Non-Symmetric Matrices)

The above equivalence fails if A is not symmetric.

Example 3.1. Let

$$A = \begin{pmatrix} 1 & 1000 \\ 0 & 1 \end{pmatrix}.$$

Then $\mathrm{eig}(A) = \{1, 1\}$, both positive, but

$$x^T Ax = x_1^2 + 1000x_1x_2 + x_2^2$$

can be negative (for example, for $x = (-1, 0.1)$). Therefore, A is not positive semidefinite. In general, a real matrix A is PSD if and only if its **symmetric part**

$$\frac{A + A^T}{2}$$

is PSD.

4 Example

Let

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The eigenvalues are $\lambda_1 = 3$, $\lambda_2 = 1$ (both positive). Therefore, A is **positive definite** since $x^T Ax > 0$ for all $x \neq 0$.

5 Summary

Property	**Eigenvalue Condition**
Positive semi-definite	$\lambda_i \geq 0$ for all i
Positive definite	$\lambda_i > 0$ for all i